

HDFS Disk Removal — High-Level Design

Remove /app8/hdfs & /app9/hdfs from all DataNodes | Ambari-managed | RHEL 7 | Azure managed disks | Replication Factor 2

Design item	Current	Target	Batch rule (RF=2)
HDFS disks per DataNode	9 (/app1/hdfs – /app9/hdfs)	7 (/app1/hdfs – /app7/hdfs)	1 DataNode preferred; 2 maximum; never 3+

PHASE 1 — PRE-CHECKS (before change, and before every batch)

1	Confirm HDFS health	hdfs dfsadmin -report — all DataNodes live; no dead nodes; cluster HEALTHY
2	Missing blocks = 0	hdfs fsck / egrep -i "MISSING CORRUPT HEALTHY" — stop and fix if any missing
3	Under-replicated = 0	hdfs dfsadmin -report grep -i "Under replicated" — must be 0 before starting

PHASE 2 — AMBARI CONFIG CHANGE (once, globally)

4	Update dfs.datanode.data.dir	HDFS → Configs → Advanced hdfs-site: remove /app8/hdfs & /app9/hdfs entries. Save. Takes effect per node only at its restart — do NOT 'Restart All Required'
---	------------------------------	--

PHASE 3 — PER-BATCH EXECUTION CYCLE (repeat until all nodes done)

5	Pick a small batch	Pilot: 1 DataNode. Normal: 1 preferred. Maximum: 2 (RF=2 rule)
6	Restart DataNode only	Ambari → Hosts → <host> → DataNode → Restart. Not the full HDFS service; NodeManager keeps running
7	Confirm DataNode healthy	Live in dfsadmin -report; no disk errors; no critical Ambari alerts
8	Confirm old paths inactive	hdfs getconf -confKey dfs.datanode.data.dir — no /app8/hdfs or /app9/hdfs; DataNode log clean
9	Let HDFS re-replicate	Blocks from removed disks re-replicate automatically from remaining replicas (only 1 copy left until done — RF=2)
10	Wait for URB = 0	GATE: under-replicated = 0, missing = 0, corrupt = 0 — do not proceed otherwise
11	Remove disks from RHEL	lsblk check → umount /app8/hdfs /app9/hdfs → backup + edit /etc/fstab → mount -a → kernel device delete (verify device via lsblk/blkid/LUN, never /dev/sdX alone)
12	Detach disks from Azure	Portal or az vm disk detach → verify with az vm show. Keep detached disks 7–14 days for rollback, then delete

GOLDEN RULES

Never >2 DataNodes per batch (RF=2) · Never unmount before the DataNode restarts with new config · Never continue while under-replicated > 0 · Never delete old block data or Azure disks until the rollback window passes

ROLLBACK

Ambari: Configs → Versions → Make Current (pre-change version) → restart DataNode · OS: restore fstab backup, remount · Azure: re-attach disk at same LUN. Old blocks re-register at next block report; excess replicas are trimmed

Full commands, checklists, and validation: [HDFS Disk Removal Procedure document](#).